

Table 1: Costs for the controlled point addition stage in Shor’s algorithm

n	Source	Strategy	#Toffoli	#CNOT	Width	Toffoli depth	Total gates	DW-cost	TDW-cost
163	[JSB+25]	In-place	63420112	793065296	53134	15088	1.62×2^{30}	1.49×2^{36}	1.49×2^{29}
		Out-of-place	31870448	426582038	15847322	7872	1.69×2^{29}	1.77×2^{43}	1.82×2^{36}
	ours	Balanced	30464596	265802254	873913	1104	1.35×2^{29}	1.88×2^{35}	1.80×2^{29}
		<i>T</i> -optimal	3167396	571478060	1643885	322	1.15×2^{29}	1.40×2^{38}	1.97×2^{28}
		Balanced+AC-inv	6341768	857859900	512450	810	1.78×2^{29}	1.08×2^{38}	1.55×2^{28}
		In-place+AC-mul	13182320	2419139728	11362	15088	1.22×2^{31}	1.77×2^{36}	1.28×2^{27}
In-place+AC-inv	13182320	2317467600	13174	13776	1.17×2^{31}	1.88×2^{36}	1.35×2^{27}		
233	[JSB+25]	In-place	142257960	2025013536	82899	23400	1.94×2^{31}	1.01×2^{38}	1.81×2^{30}
		Out-of-place	71456112	1084776730	35588718	12168	1.00×2^{31}	1.70×2^{45}	1.58×2^{38}
	ours	Balanced	62827008	585092032	1464127	1436	1.42×2^{30}	1.93×2^{36}	1.96×2^{30}
		<i>T</i> -optimal	6649524	1699691740	3443599	326	1.68×2^{30}	1.97×2^{39}	1.05×2^{30}
		Balanced+AC-inv	13394252	2570641706	916826	990	1.29×2^{31}	1.56×2^{39}	1.69×2^{29}
		In-place+AC-mul	30342312	7937855640	18133	23400	1.95×2^{32}	1.47×2^{38}	1.58×2^{28}
In-place+AC-inv	30342312	7694053848	20815	19656	1.90×2^{32}	1.42×2^{38}	1.52×2^{28}		
283	[JSB+25]	In-place	303744816	4296239536	144672	30672	1.03×2^{33}	1.67×2^{39}	1.03×2^{32}
		Out-of-place	152354640	2280405830	75930486	15904	1.06×2^{32}	1.78×2^{47}	1.10×2^{40}
	ours	Balanced	124665764	1118400118	2223915	1856	1.39×2^{31}	1.29×2^{38}	1.92×2^{31}
		<i>T</i> -optimal	10045928	3108752428	5196325	390	1.52×2^{31}	1.08×2^{41}	1.89×2^{30}
		Balanced+AC-inv	20232912	4693744666	1279002	1136	1.16×2^{32}	1.52×2^{40}	1.35×2^{30}
		In-place+AC-mul	49587536	15814797872	24202	30672	1.93×2^{33}	1.54×2^{39}	1.38×2^{29}
In-place+AC-inv	49587536	15313955920	27538	26128	1.87×2^{33}	1.52×2^{39}	1.34×2^{29}		
571	[JSB+25]	In-place	2140883888	34045115104	500450	70928	1.93×2^{35}	1.55×2^{43}	1.03×2^{35}
		Out-of-place	1072401616	17923577670	535201622	36608	1.98×2^{34}	1.72×2^{52}	1.11×2^{44}
	ours	Balanced	743713100	6657405276	12721112	2086	1.04×2^{34}	1.53×2^{41}	1.54×2^{34}
		<i>T</i> -optimal	42885298	27154260044	22123284	474	1.62×2^{34}	1.38×2^{44}	1.22×2^{33}
		Balanced+AC-inv	86049284	40693341630	5402040	1352	1.22×2^{35}	1.89×2^{43}	1.70×2^{32}
		In-place+AC-mul	246282608	162476593136	58818	70928	1.21×2^{37}	1.07×2^{43}	1.94×2^{31}
In-place+AC-inv	246282608	158325026288	65956	57200	1.18×2^{37}	1.99×2^{42}	1.76×2^{31}		